

Assignment 5

Econometrics and Machine Learning for Credit Risk Modelling

Assignment Overview

This assignment applies econometrics and machine learning to credit-risk modelling.

You will build a borrower-level default prediction system, compare econometric and machine learning models, evaluate fairness and financial-inclusion tradeoffs, and design a simple lending-policy simulation.

The assignment is not only about maximizing predictive accuracy. Your analysis must also explain:

- which borrowers are predicted to be risky,
- why the model makes those predictions,
- whether the model excludes certain groups,
- whether the lending policy is explainable and defensible,
- what limitations remain before real-world deployment.

Research Question

Can econometric and machine learning models improve credit-risk decision-making while remaining interpretable, fair, and policy-relevant?

Dataset

Use one of the following datasets.

Recommended Dataset

Home Credit Default Risk Dataset

This is the preferred dataset because it contains realistic borrower-level information, missing values, class imbalance, and a binary default target.

Alternative Dataset

Give Me Some Credit Dataset

Use this if computing resources are limited.

Learning Objectives

By the end of this assignment, you should be able to:

1. clean and analyze borrower-level credit data,
2. estimate and interpret econometric credit-risk models,
3. train and evaluate machine learning models for default prediction,
4. compare interpretability and predictive performance,
5. evaluate fairness and financial-inclusion tradeoffs,
6. design policy simulations using predicted default probabilities,
7. communicate findings through a structured analytical report.

Required Tools

Use Python.

Required libraries:

pandas
numpy
matplotlib
seaborn
scikit-learn
statsmodels
xgboost or lightgbm
shap

Optional libraries:

lifelines
econml
causalml
optuna
catboost

Part A — Data Preparation and Exploratory Data Analysis

A1. Data Summary

Report:

- number of observations,
- number of variables,
- target variable definition,
- default rate,
- variable types,
- missing-value patterns.

A2. Missing Values and Data Quality

Analyze:

- variables with high missingness,
- whether missingness appears random or systematic,
- whether missingness may contain predictive information,
- whether variables should be dropped, imputed, or transformed.

A3. Distributional Analysis

Create and interpret:

- histograms of key numeric variables,
- boxplots for income, loan amount, age, or employment duration,
- default-rate comparisons across borrower groups,
- correlation heatmap for selected variables.

A4. Class Imbalance

Discuss:

- why accuracy may be misleading,
- which metrics are more appropriate,
- how imbalance affects model training and interpretation.

Expected Output

A short EDA section with figures and written interpretation.

Part B — Econometric Credit-Risk Models

The dependent variable is the binary default indicator.

Estimate at least two of the following:

1. Linear Probability Model,
2. Logistic Regression,
3. Probit Regression.

B1. Model Specification

Define your model clearly:

$$Default_i = f(income_i, loan_i, employment_i, age_i, education_i, credit_history_i, \dots)$$

Explain why each major variable is included.

B2. Estimation and Interpretation

Report:

- coefficients,
- standard errors,
- statistical significance,
- marginal effects,
- odds ratios for logistic regression,
- economic interpretation of key variables.

Answer:

- Which borrower characteristics are associated with higher default risk?
- Are the coefficient signs economically reasonable?
- Which variables are statistically significant?
- Which variables are practically important?

B3. Econometric Concerns

Discuss:

- omitted variable bias,
- multicollinearity,
- selection bias,
- correlation versus causation,
- proxy variables for socioeconomic status.

Expected Output

A model-results table and written interpretation.

Part C — Machine Learning Models

Train at least three models:

1. regularized logistic regression using Lasso or Ridge,
2. Random Forest,
3. XGBoost or LightGBM.

C1. Preprocessing Pipeline

Describe:

- train-test split,
- missing-value imputation,
- categorical encoding,
- scaling where required,
- class imbalance handling,
- leakage prevention.

If time information is available, discuss whether a time-based split is more realistic than a random split.

C2. Model Training

For each model, report:

- main hyperparameters,
- validation method,
- tuning performed,
- final selected model.

C3. Model Evaluation

Evaluate using:

- ROC-AUC,
- PR-AUC,
- precision,
- recall,
- F1-score,
- confusion matrix,
- calibration curve,
- Brier score.

Do not rely only on accuracy.

C4. Model Comparison

Discuss:

- which model predicts best,
- which model is most interpretable,
- which model is most useful for policy,
- whether the most accurate model is necessarily the best model to deploy.

Expected Output

A model-comparison table and discussion of performance, calibration, and tradeoffs.

Part D — Explainability and Model Interpretation

Use at least two of the following:

1. feature importance,
2. permutation importance,
3. SHAP summary plot,
4. SHAP dependence plot,
5. partial dependence plot.

Required Discussion

Answer:

- Which variables are most important for prediction?
- Do the important variables make economic sense?
- Are some features proxies for income, gender, class, geography, or exclusion?
- Do ML models identify patterns that econometric models miss?
- Are these patterns interpretable enough for lending decisions?

Part E — Fairness and Financial Inclusion Analysis

Define at least two borrower groups using available variables.

Examples:

- income group,
- gender,
- age group,
- employment type,
- education level,
- family status,
- region.

Calculate at least two metrics:

1. approval rate by group,
2. default rate by group,
3. false positive rate by group,
4. false negative rate by group,
5. recall by group,
6. calibration by group.

Required Discussion

Answer:

- Which groups receive fewer approvals?
- Which groups are more likely to be incorrectly rejected?
- Does the model improve or worsen financial inclusion?
- Is there a tradeoff between predictive accuracy and fairness?
- Should all groups use the same approval threshold?

Part F — Lending Policy Simulation

Compare three lending policies:

1. Rule-Based Policy,
2. Econometric Policy,
3. Machine Learning Policy.

F1. Policy Design

Define an approval rule:

$$Approve_i = \begin{cases} 1, & \hat{p}_i(default) < threshold \\ 0, & otherwise \end{cases}$$

Test at least three thresholds, for example:

- 5%,
- 10%,
- 20%.

Alternatively, use approval-rate targets such as approving the lowest-risk 50%, 60%, or 70% of borrowers.

F2. Policy Metrics

For each policy, calculate:

- approval rate,
- predicted default rate among approved borrowers,
- false approvals,
- false rejections,
- inclusion rate for low-income borrowers,
- expected loss,

- expected repayment success.

You may use:

$$\textit{Expected Loss} = \textit{Probability of Default} \times \textit{Loan Amount} \times \textit{Loss Given Default}$$

Assume a fixed loss-given-default if the dataset does not provide one.

F3. Policy Comparison

Discuss:

- which policy reduces default most,
- which policy approves the most borrowers,
- which policy is most inclusive,
- which policy is easiest to explain,
- which policy is most defensible for regulatory use.

Part G — Causal Reasoning Extension

Choose one hypothetical intervention:

- SMS repayment reminder,
- reduced interest rate,
- repayment grace period,
- financial-literacy message,
- flexible repayment schedule.

Define:

- treatment group,
- control group,
- outcome variable,
- possible confounders.

Required Discussion

Answer:

- What is the causal question?
- What assumptions are needed to estimate the treatment effect?
- Why might treatment assignment be biased?
- What variables would you control for?
- What would make the causal estimate more credible?

Optional methods:

- matching,
- inverse probability weighting,
- doubly robust estimation,
- causal forest.

Part H — Final Recommendation Report

Write a final report of 8–15 pages.

Your report should include:

1. Introduction
2. Dataset Description
3. Exploratory Data Analysis
4. Econometric Modelling
5. Machine Learning Modelling
6. Explainability and Fairness
7. Policy Simulation
8. Causal Reasoning
9. Final Recommendation
10. Limitations

The final recommendation must answer:

- Which model should the lender use?
- Should the lender prioritize accuracy, interpretability, fairness, or inclusion?
- Which borrower groups require special attention?
- Should ML replace rule-based credit scoring?
- What safeguards are needed before deployment?
- What are the main limitations of the analysis?

Submission Requirements

Submit:

1. Final report as PDF,
2. Jupyter notebook or notebooks,
3. source code folder,
4. `requirements.txt` or `environment.yml`,
5. `README.md`,
6. figures folder,
7. saved metrics or model-output tables.

Recommended project structure:

```
assignment_5_credit_risk/  
|  
data/  
    README.md  
|  
notebooks/  
    01_eda.ipynb  
    02_econometrics.ipynb  
    03_ml_models.ipynb  
    04_policy_simulation.ipynb  
|  
src/  
    preprocessing.py  
    evaluation.py  
    fairness.py  
    policy.py  
|  
figures/  
reports/  
    final_report.pdf  
|  
requirements.txt  
README.md
```

Marking Rubric

A. Data Understanding and EDA — 15 marks

Criteria	Marks
Dataset and target-variable description	3
Missing-value and data-quality analysis	3
Distribution and imbalance analysis	3
Relevant visualizations	3
Interpretation of risk patterns	3

B. Econometric Modelling — 20 marks

Criteria	Marks
Correct use of Logit / Probit / LPM	5
Coefficient and marginal-effect interpretation	5
Statistical significance discussion	3
Bias, selection, and causality discussion	4
Calibration and policy relevance	3

C. Machine Learning Modelling — 20 marks

Criteria	Marks
Preprocessing pipeline	4
Model training and validation	4
Use of multiple models	4
Appropriate evaluation metrics	4
Model comparison and interpretation	4

D. Explainability and Fairness — 15 marks

E. Policy Simulation — 15 marks

F. Causal Reasoning — 10 marks

G. Report Quality and Reproducibility — 5 marks

Criteria	Marks
Feature importance or SHAP analysis	4
Interpretation of ML predictions	3
Fairness metrics by group	4
Financial-inclusion discussion	2
Tradeoff analysis	2

Criteria	Marks
Clear policy design	3
Comparison of rule-based, econometric, and ML policies	4
Threshold and approval-rate analysis	3
Expected loss / repayment analysis	3
Practical recommendation	2

Criteria	Marks
Clear causal question	2
Treatment/control definition	2
Confounding and identification discussion	3
Credible limitations	2
Optional estimation quality	1

Criteria	Marks
Clear writing and structure	2
Reproducible code and README	2
Clean figures and tables	1

Minimum Requirements for Passing

A passing submission must include:

- one econometric model,
- two machine learning models,
- one fairness comparison,
- one policy simulation,
- one final written recommendation,
- reproducible code.

Submissions that only train ML models without interpretation or policy analysis will not receive high marks.

Advanced Extensions

Optional extensions for stronger submissions:

- survival analysis for time-to-default,
- causal forest for heterogeneous treatment effects,
- model monitoring and drift simulation,
- Streamlit dashboard,
- FastAPI prediction API,
- threshold optimization under fairness constraints,
- cost-sensitive learning,
- reject-inference discussion,
- random split vs time-based split comparison.

Final Evaluation Focus

Your final submission should not simply report the model with the highest AUC.

A strong submission should explain:

- how different models change lending decisions,
- who benefits from each policy,
- who is excluded,
- what risks remain,
- whether the selected model is suitable for responsible deployment.

The final recommendation must balance prediction, economic reasoning, fairness, policy judgement, and reproducibility.